

GENERAL RELATIONS AMONG TRANSMISSION LINE PARAMETERS file: general.mcd

Conversion formulas included in this spreadsheet:

Inductance and capacitance to impedance	Z0()
Inductance and capacitance to propagation delay	PDLY1()
Effective permittivity to propagation delay	PDLY2()
Impedance and propagation delay to capacitance	CPI()
Impedance and propagation delay to inductance	LPI()

Variables used:

lpi	Inductance per inch (H)
cpi	Capacitance per inch (F)
pdly	Propagation delay (s/in.)
z0	Line impedance (Ω)
eeff	Effective relative permittivity

Given inductance per inch and capacitance per inch,
find the characteristic impedance in ohms:

$$Z0(lpi, cpi) := \sqrt{\frac{lpi}{cpi}}$$

Given inductance per inch and capacitance per inch,
find the propagation delay per inch:

$$PDLY1(lpi, cpi) := \sqrt{lpi \cdot cpi}$$

Given the effective electric permittivity of the surrounding medium,
find the propagation delay per inch:

$$PDLY2(eeff) := 84.72 \cdot 10^{-12} \cdot \sqrt{eeff}$$

Given impedance and propagation delay, find the capacitance per inch:

$$CPI(z0, pdly) := \frac{pdly}{z0}$$

Given impedance and propagation delay, find the inductance per inch:

$$LPI(z0, pdly) := z0 \cdot pdly$$